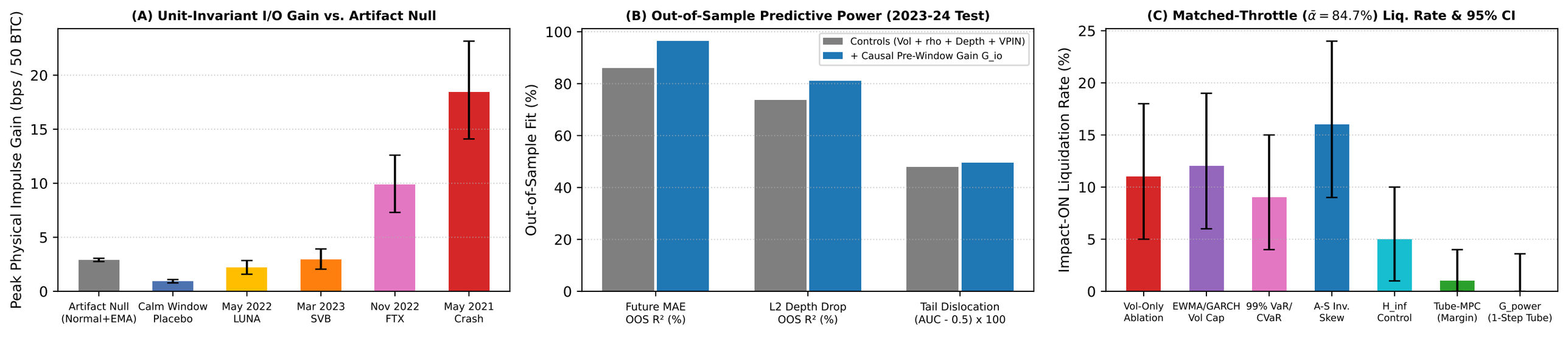
(A) Unit-Invariant I/O Gain vs. Artifact Null
Peak Physical Impulse Gain (bps / 50 BTC)
0
5
10
15
20
Artifact Null (Normal+EMA)
Calm Window Placebo
May 2022 LUNA
Mar 2023 SVB
Nov 2022 FTX
May 2021 Crash
(B) Out-of-Sample Predictive Power (2023-24 Test)
Out-of-Sample Fit (%)
0
20
40
60
80
100
Controls (Vol + rho + Depth + VPIN)
+ Causal Pre-Window Gain G_io
Future MAE OOS R² (%)
L2 Depth Drop OOS R² (%)
Tail Dislocation (AUC - 0.5) x 100
(C) Matched-Throttle ($\bar{\alpha} = 84.7\%$) Liq. Rate & 95% CI
Impact-ON Liquidation Rate (%)
0
5
10
15
20
25
Vol-Only Ablation
EWMA/GARCH Vol Cap
99% VaR/ CVaR
A-S Inv. Skew
H_inf Control
Tube-MPC (Margin)
G_power (1-Step Tube)